c) $\Pi_A = (80 - 0.5\cdot(q_1+q_2))\cdot q_1 - 4\cdot q_1 - 1200$

$\Pi_A' = 80 - q_1 - 0.5q_2 - 4 = 0$

$76 = q_1 + 0.5q_2$

$76 - 0.5q_2 = q_1$

$\Pi_F': 74 - 0.5q_1 = q_2$

$q_1 = 76 - 0.5\cdot(72 - 0.5q_1)$

$q_1 = 76 - 36 + \frac{1}{4}q_1$

$\frac{3}{4}q_1 = 40$

$q_1 = 53.\overline{3}$ $\Pi = 169$

$q_2 = 47.\overline{3}$ $\Pi = -175$

d) $\Pi_F: (80 - 0.5\cdot(q_1+q_2))\cdot q_2 - 8\cdot q_2 - 1200$

$\Pi' = 72 - 0.5q_1$

Arbeitsblatt 1:

① $C = 72 + 4q_i + 2q_i^2$

$MC = 4 + 4q_i$

b) $P = 16$

$\Pi' = 16 - 4 - 4\cdot q = 0$

$12 = 4q$

$q = 3$

$\Pi = -54$

c) $AC = \frac{72 + 4q_i + 2q_i^2}{q_i} = 34$

d) $AC = MC = P$

$\frac{72 + 4q_i + 2q_i^2}{q_i} = 4 + 4q_i$

$72 + 4q_i + 2q_i^2 = 4q_i + 4q_i^2$

$72 = 2q_i^2$

$36 = q_i^2$

$q_i = 6$

$MC = P$

$4 + 4\cdot q_i = 28 \leftarrow P$

② $P = 60 - 0.005Q$

$TC = 100'000 + 5Q + 0.0005Q^2$

$R = (60 - 0.005Q)\cdot Q$

$MR = 60 - 0.01Q$

$MC = 5 + 0.001Q$

$MR = MC$

$60 - 0.01Q = 5 + 0.001Q$

$55 = 0.011Q$

$Q = 5000$, $P = 35$

b) $R: (60 - 0.005Q)\cdot Q$

$MR = 60 - 0.01Q = 0$

$60 = 0.01Q$

$Q = 6000$

$P = 30$

③ $Q(P) = 40 - 2P$, $C = 2\cdot q + F$

$P = 20 - 0.5Q$

a) $\Pi = (20 - 0.5Q)\cdot Q - 2Q - F$

$\Pi' = 20 - Q - 2 = 0$

$Q = 18$

$\Pi = 162 - F$

b) $F \leq 162$

HÜ 2

1b) Lerner Index: $\frac{1}{|\varepsilon|}$

D_1: $\frac{1}{4} = 0.25$

D_2: $\frac{1}{2} = 0.5$

② a) $\frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q} = \frac{-25}{50} \cdot \frac{400}{125} = -1.6$

b) $Q = 4400$

$-4 \cdot \frac{100}{4400} = -0.091$

c) -1.6

③ $Q = 380 - 19P$

$P = \frac{380 - Q}{19}$

$R = P \cdot Q$

$MR = \frac{380 - 2Q}{19} = 0$

$20 = \frac{2}{19} Q$ $\quad Q = 190$

$R = 1900$

$P = 10$

b) $\frac{\Delta Q}{\Delta P} \cdot \frac{Q}{P} = -1$

$-19 \cdot \frac{P}{190} = -1$

$-19 \cdot P = -190$

$P = 10$

④ 900 S, 100 P) 1000 $\quad$ 7, 12 $\qquad MC = 2$

a) 12:

$\pi: 100 \cdot 12 - 2 \cdot 100 = 1000 \leftarrow KR: 0$

$\pi_2: 7 \cdot 1000 - 2 \cdot 1000 = 5000 \quad KR: 500$

b) $\pi: (900 \cdot 7) + (100 \cdot 12) - (1000 \cdot 2) = 5500$

$KR = 0$

HÜ 4

(1) $Q(P) = 4800 - 1200P \quad P = \frac{4800-Q}{1200}$

$w = 400 \quad C = 1.25 \cdot q + 100$

a) $Q = 1300$

$P = \frac{4800-1300}{1200} = 2.9167$

$\pi_w = 566.68$

$\pi_z = 1400.03$

b) $R = \frac{4800-(q_1+q_2)}{1200} \cdot q_1$ $\qquad MR = MC$

$MR = 4 - \frac{2q_1 - q_2}{1200} = 1.25$

$MR = 4229.215$

$2.75 = \frac{2q_1 - q_2}{1200}$

$3300 = 2q_1 + q_2$

$\frac{3300-q_2}{2} = q_1$

$q_1 = 1450$

c) $\pi_z' = 4 - \frac{2q_1 - q_2}{1200} - 1.25 = 0$

$FOC = \frac{3300-q_2}{2} = q_1 = q_2 = q$

$\frac{3300-q}{2} = q$

$3300 - q = 2q$

$3300 = 3q$

$q = 1100$

(2) $Q(P) = 100 - \frac{1}{3}P$, $TC = 150 + 2q_i$

$P = 300 - 3Q$

a) $\pi_1 = (300 - 3 \cdot (q_1 + q_2)) \cdot q_1 - 150 - 2q_1$

$\pi_1' = 300 - 6q_1 - 3q_2 - 2 = 0$

$298 = 6q_1 + 3q_2$

$\frac{298 - 3q_2}{6} = q_1$

b) 39.67

c) $\frac{298 - 3q_2}{6} = q_1 = q_2 = q$

$298 - 3q = 6q$

$298 = 9q$

$q_1 = q_2 = 33.11$

$Q = 66.22$

$P = 101.\overline{33}$

$\pi_1 = 3139.1474$

$\pi_2 = 3139.1474$

HÜ5

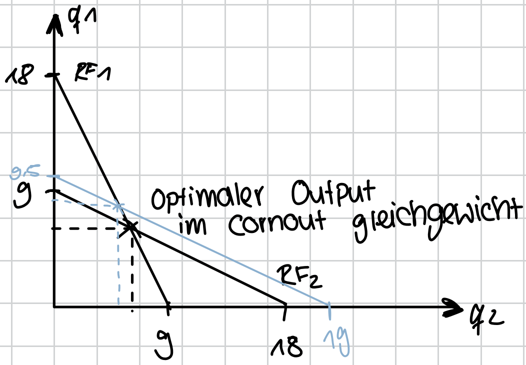
① $MC = 2$

$P(Q) = 20 - Q$

$\pi_1 = (20 - Q) \cdot q_1 - 2 \cdot q_1$

$\pi_1' = 20 - 2q_1 - q_2 - 2 = 0$

$\frac{18 - q_2}{2} = q_1 \Rightarrow q_2 = \frac{18 - q_1}{2}$



$q_1 = \frac{18 - q_2}{2}$

$q_2 = \frac{19 - q_1}{2}$

→ q_1 produziert weniger

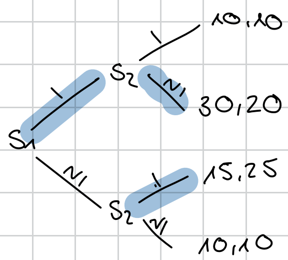
→ q_2 mehr

② a) Keine dominanten Strategien

Nash: $s_1 s_2 = (NI, I)$

$s_1 s_2 = (I, NI)$

b)



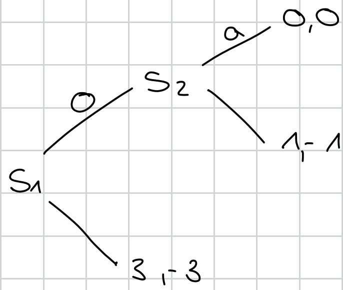
$S_1 = (I, NI)$

$S_2 = (II, INI, NII, NINI)$

c) SPNE : $s_1 s_2 = (I, NI)$

$s_1 s_2 = (NI, I)$

HÜ6



		S_2 a	b
S_1	O	0,0	1,-1
	U	3,-3	3,-3

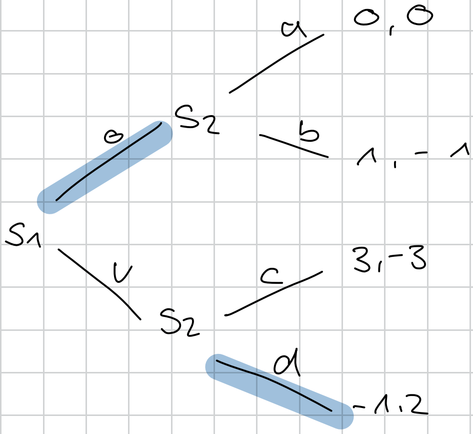
Nash- Gleichgewicht

$s_1 s_2 = (U, a)$; $s_1 s_2 = (U, b)$

SPNE

$s_1 s_2 = (U, a)$

b)



		ac	ad	bc	bd
S_1	O	0,0	0,0	1,-1	1,-1
	U	3,-3	-1,2	3,-3	-1,2

(S_2 above the columns)

Nash: $s_1 s_2 = (0, ad)$

SPNE : $s_1 s_2 = (0, ad)$

② $P = 100 - Q$

$C_1 = 10 q_1$

$C_2 = q_2^2$

$\Pi_2 = (100 - q_1 - q_2) \cdot q_2 - q_2^2$

$\Pi'_2 = 100 - q_1 - 2q_2 - 2q_2 = 0$

$\frac{100 - q_1}{4} = q_2$

$\Pi_1 = (100 - q_1 - q_2) \cdot q_1 - 10 q_1$

$= 100 q_1 - q_1^2 - \frac{100 - q_1}{4} \cdot q_1 - 10 q_1$

$100 q_1 - q_1^2 - 25 q_1 - \frac{1}{4} q_1^2 - 10 q_1$

$65 q_1 - q_1^2 - \frac{1}{4} q_1^2$

$65 q_1 - 0.75 q_1^2$

$\Pi' = 65 = 1.5 q_1$

$q_1 = 43.\overline{33}$

$q_2 = 14.167$

HÜ 7

$4 + 3\delta > 5 + \delta$

$2\delta > 1$

$\delta > 1/2$

HÜ 8

$P(Q) = 20 - 2Q \qquad MC = 2$

a) $\pi = (20 - 2 \cdot Q) \cdot Q - 2Q$

$\pi' = 20 - 4Q - 2 = 0$

$Q = 4.5$

$P = 11 \qquad \pi = 40.5 \leftarrow 20.25$

b) $P = MC = 2$

c) $\frac{20.25}{1 - \delta} > 40.5$

$20.25 > 40.5 \cdot (1 - \delta)$

$0.5 > 1 - \delta$

$\delta > 0.5$

d) $\frac{40.5 / N}{1 - \delta} > 40.5$

$\frac{1}{N} > \frac{1}{1 - \delta}$

$\delta > 1 - \frac{1}{N}$

HÜ 9

① 50 % ← 10K
5 % ← 0 $U(w) = \sqrt{w}$

$E(X) = 0.5 \cdot 10\,K = 5K$
$Eu(X) = 0.5 \cdot \sqrt{10K} + 0.5 \cdot \sqrt{0} = 50$
$S(L) \Rightarrow \sqrt{w} = 50$
$w = 2500 \leftarrow S$

Risiko:
$5K - 2.5K = 2.5K$

② 20K , 200 T $2\sqrt{w}$
$E(X) = 0.9 \cdot 20K + 0.1 \cdot 200$
$Eu(X) = 0.9 \cdot 2 \cdot \sqrt{20000} + 0.1 \cdot 2 \cdot \sqrt{200} = 257.39$
Versicherung:

$2 \cdot \sqrt{20000 - V} = 257.39$
$\sqrt{20000 - V} = 128.69$
$20\,000 - V = 16\,562$
$V = 3438$

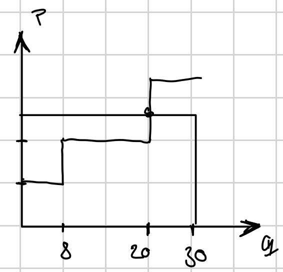
③ Autos
gut 12 700 0.6
schlecht 8 200 0.4

$\Sigma = 20$

a) g: 1200 s: 400

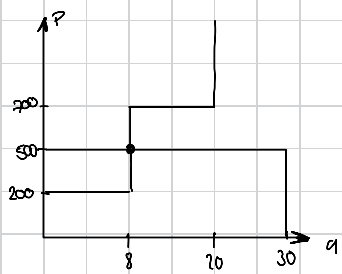
$\pi = 1200 \cdot 12 + 400 \cdot 8 \cdot 700 \cdot 20 = 3600$

b) $E(X) = 0.6 \cdot 1300 + 0.4 \cdot 500 = 980$



Alle verkauft

c) $E(X) = 500$



Nur schlecht
→ konsistent mit
E(X)

#Ü10 1000 Einwohner

$U(x, G) = x_i - \frac{100}{G}$ $x = 1€,\ G = 10€$

MRS: $m = 1000$

$U'(x) = 1$

$U'(G) = -100 \cdot G^{-1}$

$= -100 \cdot (-1) \cdot G^{-2}$

$= \frac{100}{G^2}$

$MRS = \frac{\frac{100}{G^2}}{1} = \frac{100}{G^2}$

$\hookrightarrow$ 10 € kostet 1 m² mehr Eis

b) Samuelson

$1000 \cdot \frac{100}{G^2} = 10$

$100\,000 = 10\,G^2$

$10\,000 = G^2$

$G = 100$

c) $\frac{10 \cdot G}{1000} = \frac{G}{100}$

Budgetrestriktion: $x_i + \frac{G}{100} = 1000$

$x_i = 1000 - \frac{G}{100}$

d) $U(x) = x_1 - \frac{100}{G}$

$= 1000 - \frac{G}{100} - \frac{100}{G}$

$U'(x) = -\frac{1}{100} + \frac{100}{G^2} = 0$

$\frac{100}{G^2} = \frac{1}{100}$

$100 = 0.01 G^2$

$10000 = G^2$

$G = 100$

② $U_A = x_A - (15 - y)^2$, $U_B = x_B - \frac{1}{2} \cdot (6 - y)^2$

a) A: $U'_A = 2 \cdot (15 - y) = 24$ B: $(6 - y) = 24$

$15 - y = 12$ $y = -18$ -> kauft 0

$y = 3$

b) Samuelson Bedingung

$2 \cdot (15 - y) + 6 - y = 24$

$30 - 2y + 6 - y = 24$

$36 - 3y = 24$

$3y = 12$

$y = 4$

③ $U = 2\ln x_i + \ln F \quad m = 100 \quad p = 1$

$U = 2 \cdot \ln(100 - F_1) + \ln(F_1 + F_2)$

$U' = -\frac{2}{100 - F_1} + \frac{1}{F_1 + F_2} = 0$

$\frac{1}{F_1 + F_2} = \frac{2}{100 - F_1}$

$1 = \frac{2}{100 - F_1} \cdot (F_1 + F_2)$

$100 - F_1 = 2F_1 + 2F_2$

$\frac{100 - 2F_2}{3} = F_1 \rightarrow$ symm

$\frac{100 - 2F}{3} = F$

$100 - 2F = 3F$

$100 = 5F$

$F = 20$

$F = 40$

$U = 2 \cdot \ln(95 - F_1) + \ln(F_1 + F_2 + 10)$

$U' = -\frac{2}{95 - F_1} + \frac{1}{F_1 + F_2 + 10} = 0$

$\frac{2}{95 - F_1} = \frac{1}{F_1 + F_2 + 10}$

$2F_1 + 2F_2 + 20 = 95 - F_1$

$F_1 = \frac{95 - 2F_2 - 20}{3}$

$3F = 95 - 2F - 20$

$5F = 75$

$F = 15$

$15 + 15 + 10 = 40$

↳ Crowding Out Phänomen

HÜ11

$\pi_F = 22 \cdot f - f^2 \; ; \; \pi_B = 32h - h^2 - hf$

a) $\pi = 22f - f^2 + 32h - h^2 - hf$

$\pi' = 22 - 2f - h = 0$

$11 - \frac{1}{2}h = f$

$\pi = 32 - 2h - f = 0$

$16 - \frac{1}{2}f = h$

$16 - \frac{1}{2} \cdot (11 - \frac{1}{2} \cdot h) = h$

$16 - 5.5 + \frac{1}{4}h = h$

$10.5 = \frac{3}{4}h$

$h = 14$

$f = 4$

b) $\pi_W = 22f - f^2 - fh \; , \; \pi_B = 32h - h^2$

$\pi' = 22 - 2f - h = 0 \qquad \pi' = 32 - 2h = 0$

$11 - \frac{1}{2}h = f \qquad 32 = 2h$

$f = 3 \qquad h = 16$